THE CALABI–YAU-TO-CHIRAL CORRESPONDENCE AND ITS TRACE INVARIANT

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Abstract. The first compact test is $K3$: the constructed chiral algebra has $\kappa_{\mathrm{ch}} = 2$. The product $K3 \times E$ already separates the invariants: the compact Hodge supertrace is 0, while the Heisenberg, Borcherds, and fibre readings give 3, 5, and 24. This separation fixes the correct form of the Calabi–Yau-to-chiral construction. The construction is a dimension-indexed, parameter-tagged assignment

$$\Phi_d^{(\Sigma_{d-1},C)} = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}},$$

where Φ_d^{FA} is the native holomorphic factorisation-algebra stage and $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ is the witnessed chiral specialisation. At $d = 2$ it is constructed on the smooth proper locus. At $d = 3$ a chain-level framed input produces an E_1-chiral algebra under the stated hypotheses, with the braided E_2 structure living on the Drinfeld centre of its representation category. At $d \geq 4$ the same display is a construction problem until the E_d-formality, holomorphic twist, framing, and completion witnesses are constructed.

1. Specialisation datum

The symbol $\Phi_d^{(\Sigma_{d-1},C)}$ has two different levels of meaning. On objects it is a construction from a Calabi–Yau category and a witnessed specialisation datum. On morphisms it is a functor only after the kernel, base-change, trace, and completion homotopies have been supplied.

Definition 1 (Calabi–Yau category and specialisation datum)**.** Fix a field k of characteristic zero. A *Calabi–Yau d-category* is a small idempotent-complete smooth proper k-linear dg, A_∞, or stable $(\infty, 1)$-category C, with $\mathrm{HH}^0(C) = k$, equipped with an S^1-invariant Calabi–Yau class

$$[\sigma_C] \in \mathrm{HC}^-_d(C).$$

Its image under $\mathrm{HC}^-_d(C) \to \mathrm{HH}_d(C)$ is required to be non-degenerate in the bimodule sense: the induced map from the inverse dualising bimodule to the shifted diagonal is a quasi-isomorphism. Hence cap product identifies $\mathrm{HH}^\bullet(C)$ with $\mathrm{HH}_{d-\bullet}(C)$, and the Serre functor of C is the d-fold shift. The resulting trace pairing has degree $-d$. A *witnessed specialisation datum* for C consists of a smooth Calabi–Yau d-fold X or local Calabi–Yau target with volume form, a reference curve $C \subset X$, an admissible framed $(d-1)$-cycle $\Sigma_{d-1} \subset X$, a point of the relevant formality or associator torsor, a holomorphic-twist witness, a proper-support incidence kernel, a propagator or Green-kernel normalisation, and a complete separated target in which the specialised observables converge. In dimension three the datum includes Tor-independent base-change cells, Beck–Chevalley and Fubini/envelope homotopies, compact-support conventions, and coherence data for every morphism on which specialisation is evaluated.

Definition 2 (The two stages)**.** For a Calabi–Yau d-category C with witnessed datum (X, Σ_{d-1}, C), the symbol $\Phi_d^{\mathrm{FA}}(C)$ denotes the E_d-holomorphic factorisation algebra on X assembled from (C, X, Ω_X): the Hochschild cochains of C, the chosen Kontsevich–Tamarkin formality point, Costello–Gwilliam locality, and the Costello–Li holomorphic twist with volume form Ω_X. The symbol

$$\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}(\mathcal{F}) = \left(\int_{\Sigma_{d-1}} \mathcal{F} \right)\Bigg|_C$$

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denotes factorisation-homology integration over the witnessed cycle, followed by restriction to the reference curve. A *constructed locus* is a data-bearing subcategory on which these two operations and their composite are defined. A morphism in such a locus is a cyclic dg, A_∞, or $(\infty, 1)$-functor, represented geometrically when needed by a proper Tor-independent Fourier–Mukai kernel, together with specified homotopies preserving the negative-cyclic class, target Calabi–Yau form, formality point, holomorphic twist, incidence kernel, propagator, base-change cells, compact-support convention, and completion.

Definition 3 (Three-dimensional input datum). A three-dimensional input datum is a smooth proper Calabi–Yau 3-category C over a field of characteristic zero with $\mathrm{HH}^0(C) = k$, a class $[\sigma_C] \in \mathrm{HC}_3^-(C)$ whose Hochschild shadow is non-degenerate in the preceding sense, a chain-level S^3-framing, a smooth Calabi–Yau threefold target with holomorphic volume form, a chosen formality or associator point, a holomorphic-twist witness, and a complete separated target for the resulting observables. A morphism of such data is a cyclic dg or A_∞ functor, represented on geometric inputs by a proper Tor-independent Fourier–Mukai kernel, that preserves the negative-cyclic class by a Connes-closed homotopy and also preserves the framing homotopy, formality point, holomorphic twist, proper-support incidence kernel, propagator normalisation, Beck–Chevalley and Fubini/envelope cells, compact-support convention, and completion. The witnessed specialisation datum for (Σ_2, C) is an additional chosen datum, not a consequence of the category alone.

Theorem 4 (Two-stage form). *On every constructed locus the Calabi–Yau-to-chiral object has the factorised form*

$$\Phi_d^{(\Sigma_{d-1}, C)}(C) = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1}, C}\big(\Phi_d^{\mathrm{FA}}(C)\big).$$

For $d = 2$ this gives the smooth proper Calabi–Yau 2-category construction on trace-preserving cyclic morphisms with compatible specialisation data. For $d = 3$ it gives the constructed object after fixing the witnessed datum (Σ_2, C), the three-dimensional input datum, the propagator, and the completion. It is functorial only on the data-bearing three-dimensional morphisms of the preceding definition. Functoriality on arbitrary Calabi–Yau morphisms is not asserted: cyclic kernels would first have to transport the negative-cyclic class, S^3-framing homotopy, holomorphic-twist data, proper support, Tor-independence, Beck–Chevalley cell, Fubini/envelope cell, and completion so that the two stages compose on morphisms. For $d \geq 4$ the same display names the construction problem, not a completed functor.

Proof. The first stage is assembled from the Gerstenhaber deformation complex, the chosen Kontsevich–Tamarkin formality point, Costello–Gwilliam locality, and the Costello–Li holomorphic twist in the target Calabi–Yau background. The second stage is the chiral specialisation along the chosen curve, using the specified incidence kernel, propagator, and completion. The three-dimensional input datum supplies smoothness, properness, the negative-cyclic class, and the chain-level framing. Smooth proper Calabi–Yau 2-categories need no extra three-dimensional framing datum, which is why the K3 case is the basic compact example on the constructed $d = 2$ locus. The displayed factorisation is an object-level construction unless the morphism lies in the data-bearing locus where the trace, kernel, base-change, Fubini/envelope, and completion homotopies are part of the input. On that locus the functoriality is the ordinary composition law for cyclic functors and proper Fourier–Mukai kernels, together with the Beck–Chevalley and Fubini/envelope homotopies that identify the two orders of pushforward and specialisation. □

2. Operadic level

Proposition 5 (Native operadic level). *The constructed native algebra is E_∞-chiral at $d = 1$ and E_2-chiral at $d = 2$. In every constructed $d \geq 3$ specialisation, in particular on the witnessed three-dimensional locus* *above, the specialised algebra $A = \Phi_d^{(\Sigma_{d-1},C)}(C)$ is E_1-chiral. For $d = 3$ this means $A = \Phi_3^{(\Sigma_2,C)}(C)$. Its braided E_2 enhancement is carried by*

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A)\big),$$

the Drinfeld centre of the E_1-representation category, under the dualisability and presentability hypotheses of the BZFN centre theorem. The derived bulk object $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$, the Verdier dual $A^!$, the ordered bar coalgebra $B^{\mathrm{ch}}_{E_1}(A)$, and the algebra A are distinct objects. The existence of the $d \geq 4$ output depends on the chosen higher specialisation datum and is not asserted here.

Proof. The Gerstenhaber bracket on $\mathrm{HH}^\bullet(C)$ has degree $1 - d$. After the chosen formality datum, Kontsevich–Tamarkin formality, Costello–Gwilliam locality, and the Costello–Li holomorphic twist give the native E_d-holomorphic factorisation stage on the constructed locus. Chiral specialisation is factorisation homology over Σ_{d-1}, followed by restriction to C. At $d = 1$ the Calabi–Yau target is already a curve and the commutative lattice cases refine to E_∞. At $d = 2$ the S^2-framed surface input and holomorphic curve restriction preserve the E_2-chiral structure. At constructed $d \geq 3$, Dunn–Lurie additivity and framed factorisation-homology integration over the $(d-1)$-cycle leave only the ordered E_1 direction on the reference curve. The higher Deligne theorem supplies the Hochschild and derived-centre brace structures under its stated Koszul and associator hypotheses. On constructed loci satisfying the BZFN dualisability and presentability hypotheses, the braided E_2 structure is the Drinfeld-centre structure on $\mathrm{Rep}^{E_1}(A)$. For $d \geq 4$, this is only an operadic target calculation: the chosen Σ_{d-1}, completion, and holomorphic-twist witness still have to be constructed. Global centre recovery for arbitrary Calabi–Yau 3-category objects is a separate problem. □

3. Hall positive half

Proposition 6 ($\mathbb{C}^3$ and the finite toric layer)**.** *The affine toric chart $\mathbb{C}^3$ has the spherical equivariant critical Hall positive half*

$$\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(\mathbb{C}^3) \cong Y^+(\widehat{\mathfrak{gl}}_1).$$

It is not $\mathcal{W}_{1+\infty}$. The latter belongs to the completed double and its Fock/mode realisation after the opposite half, Cartan data, Hopf pairing, radical quotient, and completion have been fixed.

For a finite toric CY_3 atlas with simplex-compatible cyclic critical DWR/Ran Rees hCS–Hall data, the local positive halves glue to the finite Rees object

$$H^+_{X_\Sigma} = \mathrm{hocolim}_{\alpha\in\Sigma(3)}\, \mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(U_\alpha).$$

This is a constructed positive-half comparison at finite Rees level. It is not a compact critical CoHA, a quasi-NCCR realisation, a Hall–Drinfeld double, or a new Φ_d-output.

Proof. The one-chart $\mathbb{C}^3$ statement is the Schiffmann–Vasserot identification of the critical Hall algebra of the three-loop Jordan quiver with potential $\mathrm{tr}(XYZ - XZY)$ with the affine Yangian positive half. The finite toric statement applies the DWR/Ran Rees hCS–Hall map on each chart and uses the face-compatible cyclic contractions and Hall wall-crossing quasi-isomorphisms on overlaps; the universal property of the finite homotopy colimit gives the displayed object. Compact Hall and doubled objects require operations not present in the positive-half colimit: monoidal vanishing-cycle realisation with compact support, continuous dualisation, non-degenerate pairing after radical quotient, Cartan data, and PBW/no-extra-relations control. □

4. Trace

Definition 7. For a compact Calabi–Yau d-fold X, meaning a smooth proper complex Calabi–Yau variety of dimension d, set

$$\Xi(X) = \sum_q (-1)^q h^{0,q}(X).$$

This is the Hodge supertrace. The symbol κ_{ch} denotes the chiral-side invariant of a constructed algebra, and is only identified with $\Xi(X)$ on the constructed Φ_d-locus.

Proposition 8 (Hodge trace on constructed loci)**.** *The scalar $\Xi(X)$ is an unconditional Hodge-theoretic invariant. If $A_X = \Phi_d^{(\Sigma_{d-1},C)}(\mathrm{Perf}(X))$ has been constructed in the compact Calabi–Yau locus together with the trace-comparison map identifying the chiral trace with the F^0-graded HKR summand, then*

$$\kappa_{\mathrm{ch}}(A_X) = \Xi(X).$$

Thus $\kappa_{\mathrm{ch}}(A_{K3}) = \Xi(K3) = 2$, and for the compact total-space output $\kappa_{\mathrm{ch}}(A_{K3\times E}) = \Xi(K3 \times E) = 0$. The categorical Euler invariant of the total space is likewise $\kappa_{\mathrm{cat}}(K3 \times E) = \chi(\mathcal{O}_{K3\times E}) = 0$; the value $\chi(\mathcal{O}_{K3}) = 2$ is a fibre value, not the product value. The other values attached to $K3 \times E$ are separate invariants:

$$\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3, \qquad \kappa_{\mathrm{BKM}} = 5, \qquad \kappa_{\mathrm{fiber}} = 24.$$

Proof. HKR decomposes Hochschild homology into the summands $\bigoplus_{p,q} H^q(X, \Omega_X^p)$. The negative-cyclic Calabi–Yau class is the S^1-equivariant lift of the Serre–Mukai trace. The only extra input in the displayed equality is the trace-comparison witness

$$\mathrm{tr}_{A_X}^{\mathrm{ch}} \circ \mathrm{HKR} = \sum_q (-1)^q \ \mathrm{pr}_{H^q(X,\mathcal{O}_X)} .$$

Thus the chiral trace functional defining κ_{ch} is literally the structure-sheaf column $\bigoplus_q H^q(X, \mathcal{O}_X)$ on the constructed compact trace locus. The equality with κ_{ch} is asserted only on loci where both the algebra A_X and this trace comparison have been built. For $K3 \times E$, Künneth gives

$$h^{0,\bullet}(K3 \times E) = (1,1,1,1), \qquad \Xi(K3 \times E) = 1 - 1 + 1 - 1 = 0.$$

The Heisenberg and Borcherds numbers above, and BCOV coefficients when present, come from their own constructions and are not HKR consequences of this projection. □

5. Bar–cobar reconstruction and duality

Definition 9 (Bar–cobar ambient category)**.** Fix a reference curve C and a complete filtered model category M_C of E_1-chiral algebras on C, with a compatible subcategory of conilpotent E_1-chiral coalgebras. For an augmented cofibrant algebra $A \in \mathsf{M}_C$, $B_{E_1}^{\mathrm{ch}}(A)$ denotes the reduced ordered E_1-bar coalgebra and $\Omega_{E_1}^{\mathrm{ch}}$ denotes its left adjoint cobar construction, both formed inside M_C or its stated coderived/pro-completed enlargement. The notation $A^!$ denotes Verdier duality on the curve C. The notation $\mathcal{C}^\vee$ denotes a Calabi–Yau/Koszul dual category only on loci where such a dual category and its trace data have been constructed.

Proposition 10. *Let A be an augmented cofibrant E_1-chiral algebra in the fixed ambient category M_C. Assume either that the reduced ordered bar coalgebra $B_{E_1}^{\mathrm{ch}}(A)$ lies in the conilpotent coalgebra subcategory where the E_1 bar–cobar adjunction is Quillen, or that the ordered bar construction is complete and separated for a weight filtration whose associated Koszul spectral sequence converges and whose associated-graded* *counit is a quasi-isomorphism. Then the chiral E_1 bar–cobar counit is a quasi-isomorphism in that ambient category or in the named coderived/pro-completed enlargement:*

$$\Omega^{\mathrm{ch}}_{E_1} B^{\mathrm{ch}}_{E_1}(A) \longrightarrow A.$$

When $C^\vee$ exists and the specialisation datum has been transported to the dual side, the stronger compatibility

$$\Phi_d^{(\Sigma_{d-1},C)}(C^\vee) \simeq \Phi_d^{(\Sigma_{d-1},C)}(C)^!$$

requires a duality-compatible kernel theorem: the Calabi–Yau/Koszul dual must transport the negative-cyclic class, formality point, framing homotopy, holomorphic-twist witness, specialisation kernel, and completion, and the transported chiral object must identify with Verdier duality. Outside loci where those data have been constructed, this is a mirror/Koszul compatibility conjecture.

Proof. On the conilpotent coalgebra locus inside M_C, the reduced bar filtration is exhaustive and complete, so the standard chiral E_1 bar–cobar comparison is a filtered quasi-isomorphism. On the ordered-bar locus the same conclusion follows from the complete separated weight filtration and the convergence of the Koszul spectral sequence; equivalently, the comparison is taken in the coderived or pro-completed ambient category where the infinite ordered bar sums converge. The displayed Φ_d-compatibility is stronger: it identifies the Calabi–Yau dual before specialisation with Verdier duality after the tagged specialisation. The bar–cobar counit alone does not supply the transported formality point, framing homotopy, holomorphic-twist witness, or specialisation kernel, so the compatibility is a conjecture until those data are constructed. □

Remark 11 (Borcherds and Igusa normalisations). The bar–cobar reconstruction above uses only the conilpotent or weight-completed ambient category stated in the proposition. It does not use an ordered-to-symmetric averaging conductor. Borcherds' weight theorem, in the Gritsenko–Nikulin denominator normalisation where $c_N(0)$ is the constant term of the Borcherds input, supplies the proved identity

$$\kappa_{\mathrm{BKM}}(\Phi_N) = \frac{c_N(0)}{2}, \qquad N \in \{1, 2, 3, 4, 6\}.$$

For these five levels $c_N(0) = (10, 8, 6, 4, 2)$, hence $\kappa_{\mathrm{BKM}} = (5, 4, 3, 2, 1)$. No equality with the compact Hodge supertrace is asserted here. When the square of the Igusa root enters, the monic Borcherds product and the OP scalar convention are kept separate:

$$D_5 = 64^{-1}\Delta_5, \qquad \Phi_{10}^{\mathrm{un}} = \Delta_5^2, \qquad \Phi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2, \qquad Z_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}.$$

The weight-five Borcherds invariant is attached to Δ_5, not to the weight-ten square; the scalar -4096 is a normalisation conversion, not a compact Hall construction.